

CATEGORY 1: STATISTICS & CENTRAL TENDENCY (Q1–Q9)

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Q1. Dataset has mean = 50, median = 70. What can you infer about skewness?

ANSWER: This is a left (negative) skew. The mean is pulled below the median because of a few very low outlier values dragging the average down.

EXAMPLE: Suppose 9 students score around 70-75 in an exam, but 1 student scores 0 due to absence. That single 0 pulls the mean far below the median, creating a long tail on the left side of the distribution.

Q2. Same mean but different variance — which dataset is more reliable?

ANSWER: The dataset with lower variance is more reliable because its values are clustered closer to the mean, meaning less unpredictability.

EXAMPLE: Two delivery companies both average 30-minute delivery times. Company A's times range 28-32 minutes (low variance), Company B's range 10-50 minutes (high variance). Even though both average the same, Company A is far more dependable for customers.

Q3. All values multiplied by 10 — what happens to mean & SD?

ANSWER: Both mean and standard deviation scale by 10 (mean $\times$ 10, SD $\times$ 10).

Variance, being the square of SD, scales by 100.

EXAMPLE: If exam scores out of 10 (mean=6, SD=1.5) are converted to a scale out of 100 (multiply by 10), the new mean becomes 60 and new SD becomes 15. Variance goes from 2.25 to 225.

Q4. Dataset has extreme outliers — which measure will you trust?

ANSWER: Median, because it is robust to outliers, unlike the mean which gets heavily influenced by extreme values.

EXAMPLE: In a small village, 9 people earn ₹20,000/month and 1 person (a billionaire) earns ₹10 crore/month. The mean income looks huge and

misleading, but the median (₹20,000) correctly reflects the typical villager's income.

Q5. Salary data — which central tendency measure is better?

ANSWER: Median is better because salary distributions are typically right-skewed; a few very high earners inflate the mean.

EXAMPLE: In a company of 100 employees, most earn ₹40,000-60,000/month, but the CEO earns ₹50,00,000/month. The mean salary would look misleadingly high, while the median accurately represents the "typical" employee.

Q6. Right-skewed dataset: mean vs median — which is larger?

ANSWER: Mean is larger than median because the long right tail (high extreme values) pulls the mean upward more than the median.

EXAMPLE: House prices in a city — most houses cost ₹50-80 lakh, but a few luxury villas cost ₹5 crore+. These few high-priced homes pull the mean price above the median price.

Q7. How would you detect skewness without plotting?

ANSWER: Compare mean vs median directly, or compute Pearson's skewness coefficient = $3 \times (\text{mean} - \text{median}) / \text{standard deviation}$.

EXAMPLE: If mean = 80, median = 65, SD = 20, then skewness = $3 \times (80 - 65) / 20 = 2.25$, a strongly positive (right-skewed) value — confirming skew without ever drawing a histogram.

Q8. Dataset A symmetric, Dataset B skewed — how will SD differ?

ANSWER: The skewed dataset (B) typically has a larger SD because extreme values in the tail stretch the spread of the data further from the mean.

EXAMPLE: Test scores that are evenly distributed around 70 (Dataset A, symmetric) will have a smaller SD than income data where a few people earn extremely high amounts (Dataset B, skewed), which inflates the spread.

Q9. If kurtosis is very high, what does it imply about tails?

ANSWER: High kurtosis (leptokurtic distribution) implies heavy/fat tails — more extreme outliers occur than in a normal distribution.

EXAMPLE: Stock market daily returns often show high kurtosis — most days have small changes, but occasional days have huge crashes or spikes (fat tails), which a normal distribution would underestimate.

CATEGORY 2: VARIANCE, SD & DISTRIBUTIONS (Q10–Q15)

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Q10. Why is normal distribution important in data science?

ANSWER: It is the foundation of the Central Limit Theorem (CLT), many ML algorithms assume normally distributed data, and it enables statistical inference and hypothesis testing.

EXAMPLE: When building a linear regression model, residuals are assumed to be normally distributed. This assumption lets us construct confidence intervals and run t-tests on the model's coefficients.

Q11. Same range but different variance — how is this possible?

ANSWER: Range only captures the min-max spread, while variance measures how spread out ALL values are. Data can have the same extremes but very different concentration in between.

EXAMPLE: Dataset A: [0, 50, 50, 50, 100] and Dataset B: [0, 10, 50, 90, 100]. Both have a range of 100, but Dataset B has much higher variance because values are spread more evenly across the range instead of bunched in the middle.

Q12. When is IQR preferred over standard deviation?

ANSWER: IQR (Interquartile Range = $Q3 - Q1$) is preferred when data is skewed or has outliers, because it is robust and unaffected by extreme

values, unlike SD.

EXAMPLE: When analyzing housing prices with a few ultra-luxury outliers, using IQR to detect "typical" price spread avoids the outliers from distorting the spread measure the way SD would.

Q13. If standard deviation is zero, what does it imply?

ANSWER: It implies all values in the dataset are identical — there is zero spread/variability.

EXAMPLE: If every student in a class scores exactly 80/100, the SD of scores is 0, since there's no variation among the values.

Q14. High variance dataset — what risk does it indicate in business?

ANSWER: High variance signals unpredictability and inconsistency, such as unstable sales figures or unreliable model outputs.

EXAMPLE: A retail store whose monthly sales fluctuate wildly between ₹1 lakh and ₹10 lakh is much riskier to plan inventory and staffing for than a store with steady ₹5-6 lakh monthly sales, even if both average the same.

Q15. How does adding an outlier affect variance?

ANSWER: Variance increases significantly, since variance is based on the squared distance from the mean, and an outlier has a very large distance that gets squared, disproportionately inflating the result.

EXAMPLE: Adding one employee earning ₹2 crore/month to a dataset of employees earning ₹30,000-50,000/month will massively spike the variance, even though it's just one data point.

CATEGORY 3: REAL-WORLD STATISTICAL THINKING (Q16–Q25)

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Q16. Company reports avg customer rating = 4.5. What could be misleading?

ANSWER: The distribution could be skewed (many 5-star and some 1-star ratings averaging out), the number of ratings could be small (low confidence), or there could be selection bias in who chooses to rate.

EXAMPLE: A restaurant with 4.5 average from only 10 reviews is far less trustworthy than one with 4.3 average from 5,000 reviews — and the 10 reviews might be mostly friends/family.

Q17. Monthly sales data — how will you detect inconsistency?

ANSWER: Check variance over time, plot a time series to visually spot anomalies, use Z-scores to flag outlier months, and verify data entry processes for errors.

EXAMPLE: If a company's sales suddenly jump 5x in one month compared to its consistent $\pm 10\%$ monthly trend, a Z-score analysis would flag that month for investigation — it might be a data entry error or a genuine one-time bulk order.

Q18. Which measure will you use to compare income inequality?

ANSWER: Gini coefficient or percentile ratios (e.g., P90/P10), since mean or median alone don't capture the shape of the income distribution.

EXAMPLE: Two countries can have the same median income, but Country A might have a Gini coefficient of 0.25 (more equal) while Country B has 0.55 (highly unequal) — a critical difference mean/median would miss.

Q19. Why is median house price reported instead of mean?

ANSWER: Real estate prices are right-skewed; a few luxury homes inflate the mean, so median gives a more representative "typical" price.

EXAMPLE: In a neighborhood of mostly ₹60 lakh homes plus one ₹10 crore mansion, reporting the mean price would mislead buyers about what they can expect to find, while the median stays accurate.

Q20. Same avg rating but different distributions — how to decide better product?

ANSWER: Compare the variance of ratings, examine the shape of the distribution, and check for bimodality (a "love it or hate it" split).

EXAMPLE: Product A gets mostly 4-star ratings (consistent), while Product B gets a mix of 5-stars and 1-stars averaging to the same score. Product A is the safer choice since B has polarized opinions.

Q21. What happens to distribution if you apply log transformation?

ANSWER: Log transformation compresses the long right tail, making right-skewed data more symmetric and closer to a normal distribution.

EXAMPLE: Income data, which is heavily right-skewed, becomes much more bell-shaped after taking $\log(\text{income})$ — making it more suitable for linear regression which assumes normally distributed errors.

Q22. Why do we standardize data before modeling?

ANSWER: To prevent features with larger numeric scales from dominating the model, and because it's required for distance-based algorithms like KNN and SVM.

EXAMPLE: If "age" (range 18-60) and "income" (range 10,000-10,00,000) are used together in KNN without standardization, income alone would dominate the distance calculation, making age almost irrelevant.

Q23. Difference between normalization and standardization?

ANSWER: Normalization rescales data to a fixed range like $[0, 1]$, while standardization shifts data to have $\text{mean}=0$ and $\text{SD}=1$. Standardization is preferred when outliers exist since normalization is sensitive to extreme min/max values.

EXAMPLE: Pixel values in images (0-255) are typically normalized to $[0, 1]$, while features like "transaction amount" in fraud detection (with outliers) are usually standardized instead.

Q24. Data is heavily skewed — what preprocessing step would you apply?

ANSWER: Apply a log, square root, or Box-Cox transformation to reduce skewness and make the data more normally distributed.

EXAMPLE: Before feeding "house price" into a regression model, applying $\log(\text{price})$ reduces the influence of the few extremely expensive houses and improves model performance.

Q25. Why is Z-score useful in outlier detection?

ANSWER: Z-score measures how many standard deviations a data point is from the mean; values with $|Z| > 3$ are typically flagged as outliers.

EXAMPLE: If average transaction amount is ₹2,000 with SD ₹500, a transaction of ₹10,000 has a Z-score of $(10000-2000)/500 = 16$, instantly flagging it as a likely fraudulent or unusual transaction.

CATEGORY 4: PROBABILITY & BAYES THEOREM (Q26–Q35)

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Q26. $P(A)=0.6$, $P(B)=0.5$ — what is max possible $P(A \cap B)$?

ANSWER: Maximum $P(A \cap B) = \min(P(A), P(B)) = 0.5$, which occurs when B is entirely a subset of A.

EXAMPLE: If 60% of customers buy "bread" and 50% buy "butter", the maximum overlap (people buying both) is 50% — that happens only if everyone who buys butter also buys bread.

Q27. When are two events independent? Give real example.

ANSWER: Two events are independent if $P(A \cap B) = P(A) \times P(B)$ — knowing one event happened gives no information about the other.

EXAMPLE: Flipping a coin and rolling a die are independent — getting "heads" on the coin doesn't change the probability of rolling a "6" on the die.

Q28. If two events are mutually exclusive, can they be independent?

ANSWER: No (unless one has probability 0) — mutually exclusive events mean if one occurs, the other cannot, so knowing one occurred completely changes the probability of the other (to zero), violating independence.

EXAMPLE: "It is raining" and "It is sunny (clear sky)" at the exact same moment are mutually exclusive — if it's raining, the probability of it being clear/sunny is 0, so they can't be independent.

Q29. Conditional probability — when is it useful in real-world ML?

ANSWER: Used extensively in Naive Bayes classifiers, fraud detection, and medical diagnosis — essentially any time we calculate $P(\text{class} \mid \text{features})$.

EXAMPLE: In email spam filtering, we calculate $P(\text{Spam} \mid \text{contains word "lottery"})$ to decide how likely an email is spam given it contains certain words.

Q30. Explain Bayes theorem in spam detection context.

ANSWER: $P(\text{Spam} \mid \text{word})$ is proportional to $P(\text{word} \mid \text{Spam}) \times P(\text{Spam})$. We update our prior belief about an email being spam (the prior, $P(\text{Spam})$) using the likelihood of seeing a particular word in spam emails.

EXAMPLE: If 40% of all emails are spam (prior), and the word "free" appears in 70% of spam emails but only 5% of legitimate emails, Bayes theorem combines these to give a much higher posterior probability that an email containing "free" is spam.

Q31. Why do we use sampling instead of full population?

ANSWER: Due to cost, time, and feasibility constraints. Sampling is much faster and, when done correctly, is sufficient because the Central Limit Theorem ensures sample statistics reliably estimate population parameters.

EXAMPLE: A national election survey can't poll all 1 billion+ citizens — instead, a random sample of 10,000 people gives a statistically reliable

estimate of the overall population's preference.

Q32. What is sampling bias? Give real-world example.

ANSWER: Sampling bias occurs when the sample is not randomly selected, causing it to misrepresent the true population and distort results.

EXAMPLE: Conducting a survey only via an online form biases results toward tech-savvy, younger, internet-connected people, missing the opinions of elderly or rural populations without internet access.

Q33. What happens if sample size increases?

ANSWER: Standard error decreases, confidence intervals narrow, and estimates become more precise and reliable.

EXAMPLE: Estimating average height from a sample of 10 people gives a wide, unreliable confidence interval; using a sample of 10,000 people gives a much tighter, more trustworthy estimate.

Q34. Difference between parameter and statistic?

ANSWER: A parameter is the true (usually unknown) value describing the entire population, while a statistic is a value computed from a sample used to estimate the parameter.

EXAMPLE: The "true average income of all Indians" is a parameter (we can never measure it exactly), while "average income calculated from a survey of 5,000 Indians" is a statistic estimating that parameter.

Q35. Why is random sampling important?

ANSWER: It ensures every unit in the population has an equal probability of being selected, which reduces bias and makes the sample truly representative of the population.

EXAMPLE: Randomly selecting customer reviews to analyze (instead of just reading the first 50 reviews shown on a website) avoids bias toward recent or algorithmically-promoted reviews.

CATEGORY 5: HYPOTHESIS TESTING (Q36–Q45)

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Q36. What is null hypothesis in simple terms?

ANSWER: It is the default assumption of "no effect" or "no difference" between groups. We use statistical evidence to try to disprove (reject) it.

EXAMPLE: When testing a new drug, the null hypothesis is "the drug has no effect on patients" — we run a clinical trial to see if there's enough evidence to reject this claim.

Q37. Type I vs Type II error — using medical test example.

ANSWER: Type I error (α) is a false positive — concluding an effect exists when it doesn't (a healthy person diagnosed as sick). Type II error (β) is a false negative — failing to detect a real effect (a sick person diagnosed as healthy).

EXAMPLE: A COVID test that says a healthy person is infected (Type I) causes unnecessary panic/quarantine; one that says an infected person is healthy (Type II) is far more dangerous as it allows disease spread.

Q38. $p\text{-value}=0.03$, $\alpha=0.05$ — what decision will you take?

ANSWER: Reject the null hypothesis, because the $p\text{-value}$ (0.03) is less than the significance level α (0.05), meaning the result is statistically significant.

EXAMPLE: Testing whether a new website design increases click-through rate, if the test gives $p=0.03$, we conclude (at 5% significance) that the new design genuinely performs differently from the old one — not just by random chance.

Q39. Why is p-value alone not enough?

ANSWER: P-value doesn't measure effect size or practical significance.

With very large sample sizes, even tiny, practically meaningless differences can become "statistically significant."

EXAMPLE: A drug trial with 1 million participants might find a p-value of 0.001 for a blood pressure reduction of just 0.1 mmHg — statistically significant, but clinically useless.

Q40. What does 'statistically significant' actually mean?

ANSWER: It means the observed result is unlikely to have occurred purely by random chance, assuming the null hypothesis is true.

EXAMPLE: If an A/B test shows Button B converts significantly better than Button A ($p < 0.05$), it means this difference is unlikely to be a fluke of random user behavior.

Q41. What does 95% confidence interval mean logically?

ANSWER: If we repeated the same study many times (say 100 times) and constructed a confidence interval each time, approximately 95 of those 100 intervals would contain the true population parameter.

EXAMPLE: A 95% CI of [50,000 - 55,000] for average salary doesn't mean there's a 95% chance the true average falls in that range for this one study — it means the method used produces a correct range 95% of the time across repeated sampling.

Q42. If confidence level increases, what happens to interval width?

ANSWER: The interval width increases, because achieving higher confidence requires capturing a wider range of plausible values.

EXAMPLE: A 99% confidence interval for average customer spend will be wider than a 90% confidence interval from the same data, since we need to be more "sure" we've captured the true value.

Q43. Why is a wide confidence interval not desirable?

ANSWER: A wide CI indicates low precision — it gives little actionable information about where the true parameter actually lies.

EXAMPLE: If a CI for "average conversion rate increase" is [0.5%, 20%], this is too vague to make a confident business decision, compared to a tight CI like [8%, 10%].

Q44. If sample size increases, how does CI change?

ANSWER: The CI narrows, because a larger sample size reduces the standard error, leading to more precise estimates.

EXAMPLE: Estimating average delivery time from 20 orders gives a wide CI; estimating from 20,000 orders gives a much narrower, more precise CI.

Q45. When would you prefer CI over hypothesis testing?

ANSWER: When estimating the actual range/magnitude of a parameter matters more than just getting a binary "reject or fail to reject" decision.

EXAMPLE: A business wants to know "by how much will revenue increase" — a confidence interval (e.g., "revenue will increase by 5%-9%") is far more useful than a simple "yes, there is a significant increase."

CATEGORY 6: ML CONCEPTS – OVERFITTING & MODEL ISSUES (Q46–Q55)

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Q46. Can a model have 100% training accuracy and still be a bad model?

ANSWER: Yes — this is a classic sign of overfitting. The model has memorized the training data (including its noise) instead of learning generalizable patterns, so it performs poorly on new, unseen data.

EXAMPLE: A decision tree with no depth limit can grow until every training example is perfectly classified, but it ends up so specific to the training set that it fails badly on test data.

Q47. Why does adding more features sometimes make a model worse?

ANSWER: This relates to the curse of dimensionality, increased risk of overfitting, and the possibility of adding noisy or highly correlated features that confuse the model rather than help it.

EXAMPLE: Adding "customer's favorite color" as a feature to a loan default prediction model adds noise without useful signal, potentially causing the model to find spurious patterns and overfit.

Q48. Two models with same accuracy — how to decide which to use in production?

ANSWER: Compare precision/recall trade-offs, inference speed, interpretability, robustness to new data, and computational/resource cost.

EXAMPLE: A complex neural network and a simple logistic regression might both achieve 90% accuracy on fraud detection, but if the logistic regression is faster, easier to explain to regulators, and nearly as robust, it may be the better production choice.

Q49. Can a decision tree overfit? How do you prevent it?

ANSWER: Yes — decision trees are prone to overfitting. Prevent it via pruning, limiting max_depth or min_samples_leaf, or using ensemble methods like Random Forest.

EXAMPLE: An unpruned tree predicting customer churn might create a unique branch for every single customer in training data; setting max_depth=5 forces it to learn general patterns instead.

Q50. What happens to logistic regression if classes are perfectly separable?

ANSWER: This causes the "complete separation" problem, where the model's coefficients diverge toward infinity, making the model numerically unstable and unreliable.

EXAMPLE: If a feature like "credit score > 800" perfectly predicts "no

default" with zero exceptions in training data, logistic regression will try to push the coefficient for that feature to infinity, causing convergence issues.

Q51. Why does K-Means fail on non-spherical data?

ANSWER: K-Means assumes clusters are spherical (round) and have roughly equal variance, since it uses Euclidean distance to the cluster centroid.

It fails on elongated, irregular, or differently-sized/shaped clusters.

EXAMPLE: If customer segments form crescent-moon or elongated shapes in feature space, K-Means will incorrectly slice them into unnatural round groups, whereas a method like DBSCAN would handle this correctly.

Q52. Increasing trees in Random Forest — will accuracy always improve?

ANSWER: Not always — accuracy tends to plateau after a certain number of trees. More trees mainly reduce variance (not bias) and beyond a point only increase computational cost without meaningful accuracy gains.

EXAMPLE: Going from 10 to 100 trees may significantly improve accuracy, but going from 500 to 1000 trees might add minimal improvement while doubling training time.

Q53. L1 vs L2 regularization effect on features?

ANSWER: L1 (Lasso) regularization can shrink some feature coefficients exactly to zero, effectively performing feature selection (sparse models). L2 (Ridge) regularization shrinks all weights toward zero but never makes them exactly zero.

EXAMPLE: In a model with 100 features where only 10 truly matter, L1 regularization will automatically zero out the 90 irrelevant features, while L2 will just reduce all 100 features' weights without removing any.

Q54. Why is log loss preferred over accuracy for classifier training?

ANSWER: Log loss penalizes confident wrong predictions much more heavily

than uncertain wrong predictions, while accuracy only checks if the final predicted class is correct, ignoring how confident/calibrated those predictions were.

EXAMPLE: A model predicting 99% confidence for the wrong class is far worse (and penalized much more by log loss) than one predicting 51% confidence for the wrong class, even though "accuracy" would treat both as equally wrong.

Q55. Training loss ↓ but validation loss ↑ after a point — what does it mean?

ANSWER: This indicates overfitting — the model is starting to learn noise/specific patterns in the training data rather than generalizable trends. Fix with regularization, dropout, or early stopping.

EXAMPLE: While training a neural network for 100 epochs, if validation loss starts increasing after epoch 30 even as training loss keeps falling, early stopping at epoch 30 (or adding dropout layers) prevents the model from overfitting further.

CATEGORY 7: PRACTICAL DATA SCIENCE TRICKY QUESTIONS (Q56–Q65)

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Q56. Applied StandardScaler on full dataset before splitting — what hidden problem?

ANSWER: This causes data leakage — the scaler's mean/SD are computed using test data too, so test set information "leaks" into training, giving overly optimistic and biased performance results.

EXAMPLE: If you scale based on the full dataset's mean income (including test rows) before splitting, your model implicitly "knows" something about the test set's distribution before being properly evaluated on it.

Q57. Model works on test set but fails in production after one week — investigate?

ANSWER: Likely data drift or concept drift — the real-world (production)

data distribution has changed compared to the training data distribution.

EXAMPLE: A fraud detection model trained on pre-holiday-season transaction patterns may fail during the holiday season when spending behavior changes dramatically (new patterns it never saw in training).

Q58. Column with 60% null values — teammate says 'just drop it'. Do you agree?

ANSWER: Not necessarily — first check if the "missingness" itself is informative (i.e., the fact that it's missing carries a signal), and consider imputation strategies or domain knowledge before dropping.

EXAMPLE: In medical data, a "test result" column being missing might mean the doctor decided the test wasn't needed (informative signal of patient health), so dropping it could remove valuable predictive information.

Q59. Used `df.fillna(df.mean())` on entire dataset — what is the mistake?

ANSWER: The mean should be computed only on the training set, not the entire (train + test) dataset. Computing it on the full dataset causes data leakage from the test set into training.

EXAMPLE: If test set values are unusually high, including them in the mean calculation inflates the fill value used in training rows, subtly biasing the model's learned patterns.

Q60. Feature importance shows one feature dominates (95%) — what do you do?

ANSWER: Investigate for potential data leakage (e.g., the feature might indirectly contain the target), validate the feature engineering process, and test model robustness/performance with that feature removed.

EXAMPLE: In a churn prediction model, if "days_since_last_login" shows 95% importance and happens to be calculated AFTER the churn event, it's likely leaking the answer rather than genuinely predicting it.

Q61. Classification model predicts only one class on imbalanced data — cause & fix?

ANSWER: The model has learned a "shortcut" of always predicting the majority class because it minimizes loss on imbalanced data. Fix using SMOTE/oversampling the minority class, setting `class_weight='balanced'`, or adjusting the decision threshold.

EXAMPLE: In a fraud dataset where only 1% of transactions are fraud, a naive model predicting "not fraud" for everything achieves 99% accuracy but is completely useless — techniques like SMOTE generate synthetic minority examples to fix this.

Q62. Merging two large datasets in Pandas crashes — what are your options?

ANSWER: Use chunking to process data in smaller pieces, use Dask for out-of-memory/parallel processing, or use SQL for efficient large joins; also optimize data types (dtypes) to reduce memory usage.

EXAMPLE: Merging two 50GB CSV files in plain Pandas may crash due to RAM limits — loading and joining them in chunks of 1 million rows at a time, or pushing the join into a SQL database, solves the memory issue.

Q63. Stakeholder asks why model predicted X for specific customer — how to answer?

ANSWER: Use explainability techniques like SHAP values or LIME to show which specific features contributed (and by how much) to that individual prediction.

EXAMPLE: For a loan rejection, SHAP values could show that "low credit score" contributed -0.4 and "high existing debt" contributed -0.3 to the prediction, giving the stakeholder (and customer) a transparent reason.

Q64. Timestamps in different time zones — problems and fix?

ANSWER: Mixed time zones break aggregations, comparisons, and sorting of events. Fix by converting all timestamps to UTC first for storage/processing, then converting to local time only for display purposes.

EXAMPLE: If user events from India (IST) and the US (EST) are stored in local time without conversion, calculating "events per hour" globally

would be incorrect — converting everything to UTC first fixes this.

Q65. Imputer seeing test data during cross-validation — what went wrong?

ANSWER: The imputer was applied outside the cross-validation pipeline (fit on the whole dataset before splitting). Fix by wrapping the imputer inside an sklearn Pipeline so it's fit only on each training fold.

EXAMPLE: Using Pipeline([('imputer', SimpleImputer()), ('model', model)]) inside cross_val_score ensures the imputer's statistics (like mean) are recalculated fresh for each fold's training data only, avoiding leakage.

CATEGORY 8: A/B TESTING & APPLIED STATISTICAL QUESTIONS (Q66–Q75)

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Q66. A/B test: small improvement but high p-value — what will you do?

ANSWER: Increase the sample size or extend the test duration. A high p-value means there isn't enough evidence yet to confirm the difference is real — it does NOT prove that variant B is worse or that there's no effect.

EXAMPLE: If Button B shows a 2% conversion improvement over Button A but $p=0.25$ with only 500 users tested, running the test for another two weeks with thousands more users may reveal a statistically significant result.

Q67. Accuracy improved from 85% to 86% — how to check if improvement is real?

ANSWER: Run a statistical significance test such as McNemar's test (comparing two classifiers on the same test set) or use cross-validation across multiple folds to confirm the improvement isn't just random noise.

EXAMPLE: Comparing Model A (85%) and Model B (86%) on the exact same 1,000 test samples using McNemar's test tells you whether the 1% gain reflects a genuinely different error pattern or just random variation.

Q68. Feature shows correlation with target — does it mean causation?

ANSWER: No — correlation does not imply causation. A third confounding variable might be influencing both the feature and the target simultaneously.

EXAMPLE: Ice cream sales correlate with drowning incidents, but ice cream doesn't cause drowning — the confounding variable is "hot summer weather," which independently increases both ice cream sales and swimming (and thus drowning risk).

Q69. Imbalanced dataset — how will statistical thinking help?

ANSWER: Use metrics like precision, recall, F1-score, and AUC-ROC instead of plain accuracy, and consider incorporating class priors into a Bayesian framing of the problem.

EXAMPLE: For a rare disease detection model where only 2% of patients are positive, reporting "98% accuracy" is meaningless — recall and precision reveal whether the model is actually catching the positive cases.

Q70. Why understanding distribution shift is important when deploying model?

ANSWER: If the input data distribution changes after deployment, model predictions degrade since the model was trained on a different distribution. Monitor using metrics like KL-divergence or Population Stability Index (PSI).

EXAMPLE: A demand forecasting model trained on pre-pandemic shopping behavior would perform poorly post-pandemic, since customer behavior distribution shifted significantly — monitoring PSI would have flagged this shift early.

Q71. Churn model has 92% accuracy but business says it is useless — why?

ANSWER: If 92% of customers naturally don't churn, a trivial model that always predicts "no churn" would also achieve 92% accuracy without identifying a single at-risk customer — making the metric meaningless for

the actual business goal.

EXAMPLE: This is exactly why recall/precision on the "churn" class (the minority class) matters far more than overall accuracy for this kind of imbalanced business problem.

Q72. Fraud model was working; now missing most fraud cases. No code changed. Why?

ANSWER: This is concept drift — fraudsters have changed their tactics/behavior over time, so the patterns the model learned no longer match current fraud patterns. The model needs retraining on recent data.

EXAMPLE: A model trained to detect fraud based on "large single transactions" becomes ineffective once fraudsters shift to many small transactions to stay under the radar — the underlying relationship between features and "fraud" has changed.

Q73. Sales forecasting breaks down every December — root cause and fix?

ANSWER: The root cause is unmodeled seasonality. Fix by adding seasonal features (month, holiday indicators), using seasonal decomposition techniques, or retraining the model with historical holiday-period data.

EXAMPLE: A simple linear regression forecasting model that ignores "holiday season" will consistently underpredict December sales every year; adding a "is_holiday_month" feature or using SARIMA (seasonal ARIMA) captures this recurring pattern.

Q74. P-value = 0.02 — what is the correct meaning?

ANSWER: If the null hypothesis (H_0) were actually true, there is only a 2% chance of observing a result this extreme (or more extreme) purely by random chance. It does NOT mean there's a 2% chance that H_0 itself is true.

EXAMPLE: A common misinterpretation is "there's a 2% chance our result is due to luck" — the correct interpretation is "assuming there's truly no effect, results this extreme would only happen 2% of the time."

Q75. Cancer detection: False Negatives are dangerous — which metric to focus on?

ANSWER: Recall (Sensitivity) = $TP / (TP + FN)$, since it specifically measures how many actual cancer cases the model successfully catches, minimizing dangerous missed diagnoses (false negatives).

EXAMPLE: A cancer screening model with high precision but low recall might rarely raise false alarms, but if it misses 40% of real cancer cases (low recall), that's far more dangerous than having some false alarms — so recall is prioritized over precision in this scenario.

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